EXPORT EDGE: AI-Powered Fruit Screening for Global Markets

Final Year Project Proposal

by

Hannan Yousaf Butt - 405326

Muhammad Moosa - 405243

Muhammad Ammar bin Akram - 414563

Advisor: Dr. Muhammad Shahzad Younis

Co-Advisor: Dr. Momina Moetesum

Department of Computing

School of Electrical Engineering and Computer Science

National University of Sciences and Technology

Islamabad, Pakistan

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1 ABSTRACT

Mango is the leading exported crop of Pakistan, yet after harvesting, and real-time export intelligence delays profitability and competitiveness in global markets. The proposed Final Year Project is an agri-tech solution called ExportEdge, which aims at transforming the mango exporting business in Pakistan. This system consists of two modules, which are integrated together: the Vision-Based Packhouse Automation System and the Multi-modal Export Intelligence Platform. The former is the first module that will utilize computer vision and deep learning to automate post-harvest sorting based on grading. It undertakes variety classification and sorting based on ripeness, size, and flaws and produces high-quality produce in good quantities across all export markets. The second module is an Artificial Intelligence application that allows exporters to key in an image of a graded mango and receive information on current market demand trends in the global market, competitive pricing, and country-related regulations compliance in real-time. Bridging the visual and physical quality measurement gap and online market, ExportEdge will help reduce post-harvest loss, enhance product stability, and empower Pakistani farmers and exporters to make informed, lucrative decisions. Besides practical use, this project applies to the realm of smart farming by combining the quality assurance of innovations in AI with the export intelligence of the data, which will contribute to a more prominent Pakistani niche in the international market.

2 INTRODUCTION

Pakistan is the world's 5th largest producer of mangoes, with a significant portion of its economy tied to mango cultivation and export. However, the industry faces substantial challenges, primarily in post-harvest handling and market access. Traditional manual sorting and grading methods are labor-intensive, slow, prone to human error, and result in significant post-harvest losses, often ranging from 25% to 40%. Furthermore, exporters often lack real-time market intelligence, leading to suboptimal destination choices and pricing strategies, which severely impact profitability and market share. The challenge is to create a system that addresses these inefficiencies by leveraging modern AI and automation technologies, enabling a more streamlined, data-driven, and profitable export workflow.

3 PROBLEM STATEMENT

The current mango export process is hindered by two primary bottlenecks:

1. **Inefficient and Error-Prone Post-Harvest Handling:** Manual sorting and grading lead to inconsistencies in product quality, high rates of rejection from international buyers, and substantial fruit waste. The lack of a uniform grading standard, especially for variety-specific sorting, makes it difficult to meet the precise demands of high-value markets.

2. **Lack of Actionable Market Intelligence:** Exporters often rely on outdated or anecdotal information to decide on export destinations and pricing. They lack a centralized, real-time platform that provides an analysis of global demand, competitor pricing, and complex, ever-changing import regulations. This information gap leads to missed opportunities, financial losses due to rejections, and a significant competitive disadvantage. The problem is a lack of a single, intelligent, and integrated system to address both the physical quality control and the digital market analysis aspects of the export business.

3.1 Who needs it

1. **Mango Exporters & Traders (Target Market Size: 5,000 in Pakistan):** These are the direct users who need a solution to ensure their product meets international standards, reduce rejections, and maximize profits by identifying the most lucrative markets.

2. **Large-Scale Mango Growers & Packhouse Owners (Target Market Size: 3,000 in Pakistan):** They need to automate their post-harvest processes to increase efficiency, reduce labor costs, and guarantee a high-quality product that commands a premium price.

4 LITERATURE REVIEW

Traditional computer vision systems have been used for fruit sorting based on simple metrics like size and color. However, these systems often lack the precision to identify subtle defects and, more importantly, differentiate between closely related mango varieties. Recent advancements in deep learning, particularly Convolutional Neural Networks (CNNs), have shown high accuracy in image classification and object detection tasks. Studies have demonstrated the use of CNNs for fruit quality assessment and even variety identification, but these systems are often standalone. Similarly, market intelligence platforms exist for agricultural commodities, but they are generally text-based and do not integrate with the physical product sorting process. No single solution is available in the market that combines real-time computer vision for a specific variety and quality sorting with a multi-modal AI platform for market and regulatory intelligence. This lack of a unified, end-to-end system for the mango value chain presents a significant gap that this project aims to fill.

4.1 Multi-modal LLMs for Agricultural Market Intelligence

The agri-tech sector has made significant progress in areas such as precision farming and crop monitoring; however, the application of large language models (LLMs) and multi-modal AI for export-oriented market intelligence remains limited. This review examines existing solutions, their shortcomings, and the value proposition of the proposed Export-Edge platform.

4.1.1 Existing AI Solutions for Agricultural Market Intelligence

Current systems fall into two categories:

- **Text-based platforms:** Tools such as S&P Global Commodity Insights use LLMs to process large volumes of unstructured data—market news, commodity prices, and financial reports—to provide real-time pricing, demand forecasting, and risk management.
- **Computer vision systems:** Used mainly for crop and produce assessment, focus-

ing on grading and quality evaluation.

These tools excel individually but remain disconnected from each other.

4.1.2 Disadvantages and Gaps

Several gaps make current platforms unsuitable for exporters, particularly Pakistan's mango industry:

- **Lack of Integration:** Pricing platforms and quality-control systems operate separately, forcing exporters to manually bridge the gap.
- **Absence of Multi-modal Interfaces:** Research on multi-modality has focused on farming operations (e.g., field notes, sensor data), not on market-facing tools. Exporters cannot use an image of a product as a query; instead, they rely on generic keyword searches like "mango price."
- **Generalized Data:** Existing platforms give broad commodity insights but rarely provide variety or grade-specific intelligence (e.g., Chaunsa vs. Sindhri). Exporters, however, need fine-grained, compliance-aware insights for premium markets.
- **Cost and Accessibility Barriers:** Commercial tools are expensive and technically complex, often designed for multinationals, making them inaccessible to small and medium exporters—the majority of Pakistan's mango trade.

4.1.3 ExportEdge's Differentiators

ExportEdge addresses these limitations through:

- **End-to-End Integration:** Linking mango sorting and grading directly with market intelligence, creating a unified workflow.
- **Multi-modal Querying:** Exporters can upload an image of their graded mango, and the system generates a targeted market report with pricing, compliance, and destination recommendations.

- **Granular and Contextual Insights:** Intelligence is tailored to specific varieties and grades, e.g., “Grade A Sindhri mangoes in Dubai,” including tariffs and documentation requirements—something absent from current platforms.

5 PROJECT OVERVIEW/GOAL

The overall goal of this project is to develop and implement ExportEdge, a dual-component system that bridges the physical and digital aspects of the mango export process.

Proposed Solution:

ExportEdge is composed of two integrated modules: a Multi-modal Export Intelligence Platform and a Vision-Based Packhouse System. The first module uses computer vision and deep learning to automate the process of grading mangoes, while the second module connects the physical quality results with real-time market data, price data, and regulatory compliance data. The combination will create a single platform on which the mango export process becomes a data-driven one.

Value Proposition:

Our solution is fundamentally different and superior to existing systems because it provides an end-to-end, integrated solution. It doesn’t just sort mangoes; it uses that sorting to provide actionable, data-driven insights that directly translate to increased profitability and market access. The multi-modal AI component, which takes an image as input and provides a comprehensive market analysis, is the key differentiator, offering an unprecedented level of convenience and intelligence to the exporter.

5.1 Final Project Output:

- **Hardware Component:** A physical prototype of the Vision-Based Packhouse System. This will include a conveyor belt mechanism, a high-resolution camera for grading of mangoes and an arm for sorting the mangoes.

- **Software Components:**

  - A deep learning model (e.g., a CNN) trained to identify key Pakistani mango varieties and grade them based on size, ripeness, and defects.

- A web-based platform (ExportEdge) for the Multi-modal Export Intelligence component. The platform will have a user interface for uploading mango images and a backend that processes the request and interacts with a dynamic database of market and regulatory information. The platform will use Firebase for user authentication and data storage.

- **Expected Packaging:** The final project output will be a working physical prototype of the sorting system and a fully functional, mobile-responsive web application that can be accessed from any device.

6 PROJECT DEVELOPMENT METHODOLOGY/ARCHITECTURE

The project will be developed in a modular fashion with the following key objectives and deliverables:

- **Module 1: Vision-Based Packhouse Automation System**
  - **Objective 1.1: Data Collection & Model Training:** Acquire a dataset of high-quality images of Pakistani mango varieties (e.g., Chaunsa, Sindhri, Anwar Ratol) at different stages of ripeness and with various defects. Train a CNN model for grade classification.
  - **Objective 1.2: Hardware Integration:** Build a small-scale prototype of the conveyor system with a camera that grades the mangoes.
  - **Deliverable:** A trained computer vision model and a functional hardware prototype capable of sorting mangoes into bins based on grade.
- **Module 2: Multi-modal Export Intelligence Platform**
  - **Objective 2.1: Platform Design & Frontend Development:** Design and develop the web-based user interface for the ExportEdge platform. This will include the image upload feature and display panels for market data.
  - **Objective 2.2: Backend Development & API Integration:** Develop the backend logic to handle image processing and integrate with a dynamic

knowledge base or API for real-time market data (e.g., using Google Search and other publicly available APIs to scrape data on current prices and regulations).

  - **Deliverable:** A complete, functional web application that accepts image input and provides a structured, data-driven analysis.

- **Tools and Technologies:**
  - **Deep Learning:** Python with libraries like TensorFlow/PyTorch for model training.
  - **Hardware/Embedded Systems:** Arduino or Raspberry Pi for controlling the conveyor belt and sorting mechanism.
  - **Web Development:** React or Angular for the front end, Node.js/Express.js for the backend.
  - **Database:** Firestore for storing user data and any persistent information.

7 PROJECT MILESTONES AND DELIVERABLES

- **Milestone 1 (Month 1-2):** Literature Review and Project Planning. **Deliverable:** Completed Proposal Document.
- **Milestone 2 (Month 3-4):** Vision System Development. **Deliverable:** Functional computer vision model and a hardware prototype.
- **Milestone 3 (Month 5-6):** AI Platform Development. **Deliverable:** Functional web application with a basic multi-modal AI feature.
- **Milestone 4 (Month 7-8):** Integration & Final Testing. **Deliverable:** Integrated end-to-end system and final project report.

Milestone	**Timeline**	**Deliverable**	**Description**
Milestone 1: Literature Review and Project Planning	Month 1–2	Completed Proposal Document	We will begin by conducting an in-depth literature review to understand existing research, technologies, and practices in our domain. This will help us identify gaps and opportunities. Based on our findings, we will define the project objectives, scope, and methodology. By the end of this phase, we will produce a comprehensive proposal document outlining our plan, resources, and timeline.
Milestone 2: Vision System Development	Month 3–4	Functional computer vision model and a hardware prototype	During this stage, we will focus on developing the core vision component of our system. We will design, train, and evaluate a computer vision model to meet our project requirements. Alongside, we will create a hardware prototype (such as a camera integrated with a processing unit) to test real-world feasibility and ensure seamless interaction between the hardware and the vision model.

Milestone	Timeline	Deliverable	Description
Milestone 3: AI Platform Development	Month 5–6	Functional web application with a basic multi-modal AI feature	Here, we will develop a web-based platform to host and interact with our system. We will implement a user-friendly interface and integrate basic multi-modal AI features, such as the ability to process images, text, or voice. This milestone will make our AI accessible and easy to use.
Milestone 4: Integration & Final Testing	Month 7–8	Integrated end-to-end system and final project report	Finally, we will integrate all components — the vision system, hardware prototype, and AI platform — into a fully operational solution. We will carry out thorough testing to ensure accuracy, stability, and usability. At the end of this phase, we will deliver a final report documenting our approach, results, challenges, and recommendations for future improvements.

7.1 Work Division

The project workload is distributed among team members based on their expertise and interests, with each member taking primary responsibility for specific components while collaborating on integration and testing phases.

Team Member	**Primary Responsibility**	**Detailed Tasks and Deliverables**
Muhammad Ammar bin Akram	Computer Vision System Development	**Phase 1 (Months 1-3):** • Conduct literature review on CNN architectures for fruit classification • Design and implement data collection strategy for mango varieties • Develop image preprocessing pipeline (augmentation, normalization) **Phase 2 (Months 3-5):** • Design and train CNN models for variety classification • Implement defect detection algorithms • Optimize model performance for real-time processing • Develop model validation and testing protocols **Phase 3 (Months 5-7):** • Integrate vision model with hardware components • Implement camera calibration and lighting optimization • Develop quality assurance metrics and feedback loops

Team Member	**Primary Responsibility**	**Detailed Tasks and Deliverables**
Muhammad Moosa	Large Language Model Development	**Phase 1 (Months 1-3):** • Research multi-modal LLM architectures and APIs • Design system architecture for market intelligence platform • Develop data collection strategies for market information **Phase 2 (Months 3-6):** • Implement multi-modal AI processing pipeline • Develop natural language processing for market queries • Create real-time data integration APIs for pricing and regulations • Build knowledge base for mango varieties and global markets **Phase 3 (Months 6-8):** • Implement intelligent recommendation algorithms • Develop market trend analysis and prediction features • Create automated report generation system • Optimize LLM response accuracy and speed

Team Member	Primary Responsibility	Detailed Tasks and Deliverables
Hannan Yousaf Butt	Edge Device Deployment & Web Development	**Phase 1 (Months 1-3):** • Design system architecture and technology stack • Research embedded systems for packhouse integration • Plan web application user interface and user experience **Phase 2 (Months 3-6):** • Develop responsive web application frontend (React/Angular) • Implement user authentication and authorization systems • Create dashboard for real-time monitoring and analytics • Develop mobile-responsive interface for field use **Phase 3 (Months 6-8):** • Deploy edge computing solutions on Raspberry Pi/Jetson Nano • Implement hardware integration and control systems • Develop automated sorting mechanism control software • Create system monitoring and maintenance interfaces

7.1.1 Collaborative Responsibilities

All team members will work together on the following cross-functional activities:

Activity	Timeline	Collaborative Tasks
System Integration	Months 6-7	• Integration of computer vision model with web platform • API development for communication between hardware and software components • End-to-end workflow testing and optimization • Performance benchmarking and system validation
Testing & Validation	Months 7-8	• Comprehensive system testing with real mango samples • User acceptance testing with potential exporters • Performance testing under various operational conditions • Security testing and vulnerability assessment
Documentation	Months 1-8	• Technical documentation and system manuals • User guides and training materials • Research paper writing and publication preparation • Final project report compilation
Project Management	Months 1-8	• Weekly progress meetings and milestone reviews • Risk assessment and mitigation planning • Resource allocation and timeline management • Stakeholder communication and feedback incorporation

7.1.2 Backup and Support Structure

To ensure project continuity and knowledge sharing:

- **Cross-training:** Each team member will maintain basic understanding of other components to provide backup support when needed.

- **Code Reviews:** All major developments will undergo peer review to ensure code quality and knowledge transfer.
- **Regular Integration Meetings:** Weekly technical meetings to discuss progress, challenges, and integration requirements.
- **Shared Documentation:** Maintain comprehensive shared documentation using collaborative platforms (Git repositories, shared drives).
- **Skill Development:** Team members will participate in relevant workshops, online courses, and technical conferences to enhance project-relevant skills.

7.2 Costing

- **Hardware:**
  - Conveyor Belt Motor & Materials
  - Raspberry Pi/Jetson Nano
  - High-Resolution Camera
  - Sensors and Wires
- **Software:**
  - API Subscriptions (if needed)

8 CHALLENGES AND LIMITATIONS

The development and implementation of ExportEdge present several technical, operational, and economic challenges that must be carefully addressed:

8.1 Technical Challenges

1. **Data Collection and Quality:** Acquiring a comprehensive dataset of Pakistani mango varieties with sufficient variation in ripeness stages, defects, and environmen-

tal conditions poses a significant challenge. The quality and diversity of training data directly impact the accuracy of the computer vision model.

2. **Real-time Processing Requirements:** The system must process images and provide market intelligence within acceptable time frames for commercial packhouse operations. Balancing computational complexity with processing speed while maintaining accuracy is a critical technical challenge.

3. **Hardware Integration Complexity:** Synchronizing the conveyor belt system with computer vision processing and mechanical sorting arms requires precise timing and coordination. Environmental factors such as lighting conditions, vibration, and dust in packhouse environments may affect system performance.

4. **API Reliability and Data Consistency:** The multi-modal platform depends on external APIs for real-time market data. Ensuring consistent data availability, handling API rate limits, and managing data quality from multiple sources presents ongoing technical challenges.

5. **Scalability and Performance:** As the system scales to handle larger volumes of mangoes and serve multiple exporters simultaneously, maintaining performance while managing increased computational and network loads becomes challenging.

8.2 Operational and Economic Challenges

1. **Industry Adoption Resistance:** Traditional mango exporters may be resistant to adopting new technology due to established practices, concerns about reliability, and the learning curve associated with new systems.

2. **Cost of Implementation:** Initial investment in hardware, software development, and system integration may be prohibitive for smaller exporters, potentially limiting market penetration.

3. **Maintenance and Support:** Providing ongoing technical support, system maintenance, and updates in remote or underserved areas presents logistical challenges.

4. **Market Data Accuracy:** Ensuring the accuracy and timeliness of global market data, pricing information, and regulatory compliance requirements across different countries and markets is complex and resource-intensive.

5. **Competition from Established Solutions:** Competing with existing, well-established agricultural technology providers and convincing users to switch from current systems poses market entry challenges.

8.3 Regulatory and Compliance Challenges

1. **Food Safety Standards:** The hardware system must comply with food safety regulations and hygiene standards required in packhouse operations.

2. **Export Regulation Complexity:** Different countries have varying import regulations, quality standards, and documentation requirements. Keeping the system updated with changing international trade regulations is challenging.

3. **Data Privacy and Security:** Handling sensitive business information, including pricing data and export volumes, requires robust data security measures and compliance with data protection regulations.

9 SUSTAINABLE DEVELOPMENT GOALS (SDGS) ALIGNMENT

ExportEdge directly contributes to multiple United Nations Sustainable Development Goals, demonstrating its potential for positive global impact:

9.1 Primary SDG Alignment

- **SDG 2: Zero Hunger**
  - **Target 2.3:** By improving post-harvest handling and reducing food waste by 25-40%, ExportEdge helps increase agricultural productivity and incomes of small-scale food producers, particularly mango farmers in Pakistan.

  - **Target 2.4:** The system promotes sustainable food production systems through efficient resource utilization and waste reduction in the mango supply chain.

- **SDG 8: Decent Work and Economic Growth**
  - **Target 8.2:** ExportEdge aims to achieve higher levels of economic productivity through diversification, technological upgrading, and innovation in the agricultural export sector.
  - **Target 8.3:** The platform supports development-oriented policies that promote productive activities, decent job creation, and formal sector growth by improving market access for mango exporters.
  - **Target 8.9:** By enhancing export competitiveness, the system promotes sustainable tourism and local products that create jobs and promote local culture (Pakistani mango varieties).

9.2 Secondary SDG Contributions

- **SDG 1: No Poverty**
  - **Target 1.1 & 1.2:** By reducing post-harvest losses and improving market access, ExportEdge helps increase incomes of mango farmers and exporters, contributing to poverty reduction in rural Pakistan.
- **SDG 9: Industry, Innovation and Infrastructure**
  - **Target 9.4:** The project upgrades infrastructure and retrofits industries to make them sustainable, with increased resource-use efficiency and adoption of clean and environmentally sound technologies.
  - **Target 9.5:** ExportEdge enhances scientific research and upgrades technological capabilities of Pakistan's agricultural sector through AI and automation technologies.
- **SDG 12: Responsible Consumption and Production**

  - **Target 12.3:** By reducing food losses along production and supply chains, including post-harvest losses, the system directly contributes to halving per capita global food waste.
  - **Target 12.6:** The platform encourages companies (exporters) to adopt sustainable practices and integrate sustainability information into their reporting cycle.

- **SDG 17: Partnerships for the Goals**
  - **Target 17.8:** ExportEdge promotes the use of information and communications technology to enhance capabilities and provide market intelligence to developing country exporters.
  - **Target 17.11:** The system aims to significantly increase Pakistan's exports and help diversify export markets through improved quality and market intelligence.

9.3 Quantified SDG Impact Projections

- **Food Waste Reduction:** Potential to reduce post-harvest losses by 15-25%, saving approximately 50,000-100,000 tons of mangoes annually in Pakistan.
- **Income Enhancement:** Projected 20-30% increase in export revenues for participating farmers and exporters through better quality control and market targeting.
- **Employment Creation:** Direct creation of technical jobs in system maintenance, data analysis, and customer support, plus indirect job creation through improved agricultural productivity.
- **Technology Transfer:** Knowledge and technology transfer to the agricultural sector, building local capacity in AI and automation technologies.

9.4 Long-term SDG Contributions

The successful implementation of ExportEdge creates a foundation for broader agricultural transformation:

- **Scalability:** The technology framework can be adapted for other fruits and agricultural products, multiplying its SDG impact across Pakistan's agricultural sector.
- **Regional Impact:** Success in Pakistan could lead to technology transfer and adaptation in other developing countries facing similar agricultural export challenges.
- **Digital Agricultural Ecosystem:** ExportEdge contributes to building a digital agricultural ecosystem that supports sustainable development goals through data-driven decision making and improved market access.

10 REFERENCES

REFERENCES

[1] S&P Global. "AI Powered Solutions." S&P Global Commodity Insights. Available at: https://www.spglobal.com/commodity-insights/en/products-solutions/ai

[2] Revenue AI. "Commodity Trading Industry Offer." Revenue AI. Available at: https://revenue.ai/industry-commodity-trading/

[3] AWS. "Revolutionizing agricultural knowledge management using a multi-modal LLM: A reference architecture." AWS Blog. Available at: https://aws.amazon.com/blogs/architecture/revolutionizing-agricultural-knowledge-management-using-a-multi-modal-llm-a-refere

[4] Moroccan Journal of Agricultural Sciences. "Artificial intelligence for enhancing supply chain management in agribusiness." Moroccan Journal of Agricultural Sciences. Available at: https://agromaroc.com/index.php/MJAS/article/download/1068/1190

[5] ResearchGate. "A Study on Problems and Prospects of Agricultural Marketing in the Age of Artificial Intelligence and Smart Farming." ResearchGate. Available at: https://www.researchgate.net/publication/394949304_A_Study_on_Problems_and_Prospects_of_Agricultural_Marketing_in_the_Age_of_Artificial_Intelligence_and_Smart_Farming

[6] Software House. "Disadvantages of AI in Agriculture: Key Challenges to Consider." Software House Blog. Available at: https://softwarehouse.au/blog/what-are-the-disadvantages-of-ai-in-agriculture/